

Financial valuation of actuarial liabilities
Homework 2 - Valuation of a participating annuity

Alexis Thieltgen - 60292100

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Question 1

The indice $(S_t)_{t \geq 0}$ is driven by a geometric Brownian motion with volatility σ and continuous risk-free rate r , so its dynamics is defined by

$$\frac{dS_t}{S_t} = rdt + \sigma dW_t .$$

The x -year old customer investing C_0 in the contract (or its inheritor) will receive a payment depending on his survival or not until maturity T . We therefore observe two main parts in the contract: a GMAB component (payment at maturity in case of survival of the insured) and a GMDB component (payment in case of premature death at time t).

GMAB component:

In case of survival at expiry T , the insured receives $C_0 \max(\frac{S_T}{S_0}, e^{gT})$ where g is the guaranteed rate. The contract value is the sum of expected discounted cash flows under the risk neutral measure Q . Denoting the remaining lifetime of the insured by τ , we thus have

$$V_0^{GMAB} = \mathbb{E}^Q[\mathbf{1}_{\{\tau > T\}} e^{-rT} C_0 \max(\frac{S_T}{S_0}, e^{gT}) | \mathcal{F}_0] .$$

Rewriting the payoff as

$$\begin{aligned} C_0 \max(\frac{S_T}{S_0}, e^{gT}) &= \frac{C_0}{S_0} \max(S_T, S_0 e^{gT}) \\ &= \frac{C_0}{S_0} (S_0 e^{gT} + \max(S_T - S_0 e^{gT}, 0)) \\ &= \frac{C_0}{S_0} (S_0 e^{gT} + (S_T - S_0 e^{gT})_+) , \end{aligned}$$

and using the fact that mortality is independent from financial markets, we get the following:

$$\begin{aligned} V_0^{GMAB} &= \mathbb{E}^Q[\mathbf{1}_{\{\tau > T\}} | \mathcal{F}_0] \mathbb{E}^Q[e^{-rT} \frac{C_0}{S_0} (S_0 e^{gT} + (S_T - S_0 e^{gT})_+) | \mathcal{F}_0] \\ &= {}_T p_x(e^{(g-r)T} C_0 + \frac{C_0}{S_0} \mathbb{E}^Q[e^{-rT} (S_T - S_0 e^{gT})_+ | \mathcal{F}_0]) , \end{aligned}$$

with the probability of survival ${}_T p_x = Q(\tau > T)$. Since $(S_t)_{t \geq 0}$ is a geometric Brownian motion, the expectation can be valued with the Black-Scholes formula. Setting the strike as $K := S_0 e^{gT}$, the value of the last embedded call option with strike K is

$$\mathbb{E}^Q[e^{-rT} (S_T - K)_+ | \mathcal{F}_0] = S_0 \Phi(d_1^{BS}) - K e^{-rT} \Phi(d_2^{BS})$$

where

$$d_1^{BS} = \frac{\ln \frac{S_0}{K} + (r + \frac{\sigma^2}{2})T}{\sigma \sqrt{T}} , \quad d_2^{BS} = d_1^{BS} - \sigma \sqrt{T} ,$$

and where $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

GMDB component:

Discrete version

In case of premature death at time $t \in [0, T]$, the inheritor receives $C_0 \max(\frac{S_t}{S_0}, e^{gt})$. The contract can terminate at any time before maturity and pays a value depending on when death occurs. We thus have to consider a continuum of possible payment dates. To ease calculations, we could discretize the interval $[0, T]$ and assume (just like in the lectures) that if $\tau \in [k-1, k)$, payment occurs at time $k \in \mathbb{N}$ (end of the year). Note that a continuous version of the following formulas also exists, involving integrals and the force of mortality μ_{x+t} (detailed just later).

We again depart from the sum of expected discounted cash flows:

$$V_0^{GMDB} = \mathbb{E}^Q \left[\sum_{k=1}^T \mathbf{1}_{\{k-1 \leq \tau < k\}} e^{-rk} \max\left(\frac{S_k}{S_0}, e^{gk}\right) | \mathcal{F}_0 \right].$$

Mortality being independent from financial markets, we can again split the expectation. Moreover, rewriting the max function as before yields

$$\begin{aligned} V_0^{GMDB} &= \sum_{k=1}^T \mathbb{E}^Q[\mathbf{1}_{\{k-1 \leq \tau < k\}} | \mathcal{F}_0] \mathbb{E}^Q[e^{-rk} C_0 \max\left(\frac{S_k}{S_0}, e^{gk}\right) | \mathcal{F}_0] \\ &= \sum_{k=1}^T Q(k-1 \leq \tau < k) \mathbb{E}^Q[e^{-rk} \frac{C_0}{S_0} (S_0 e^{gk} + (S_k - S_0 e^{gk})_+) | \mathcal{F}_0] \\ &= \sum_{k=1}^T Q(\tau < k | \tau \geq k-1) Q(\tau \geq k-1) (C_0 e^{(g-r)k} + \frac{C_0}{S_0} \mathbb{E}^Q[e^{-rk} (S_k - S_0 e^{gk})_+] | \mathcal{F}_0) \\ &= \sum_{k=1}^T q_{x+k-1} {}_{k-1}p_x (C_0 e^{(g-r)k} + \frac{C_0}{S_0} \mathbb{E}^Q[e^{-rk} (S_k - S_0 e^{gk})_+] | \mathcal{F}_0) \end{aligned}$$

where ${}_t q_x = 1 - {}_t p_x = Q(\tau \leq t)$ is the probability of dying before $x+t$. Let us define the strike $K(k) := S_0 e^{gk}$. The Black-Scholes formula yields

$$\mathbb{E}^Q[e^{-rk} (S_k - K(k))_+ | \mathcal{F}_0] = S_0 \Phi(d_1^{BS}(k)) - K(k) e^{-rk} \Phi(d_2^{BS}(k))$$

where

$$d_1^{BS}(k) = \frac{\ln \frac{S_0}{K(k)} + (r + \frac{\sigma^2}{2})k}{\sigma \sqrt{k}} \quad \text{and} \quad d_2^{BS}(k) = d_1^{BS}(k) - \sigma \sqrt{k}.$$

Continuous version

In the continuous time version, the payment occurs directly at the moment the insured dies, $\tau = t$. So instead of a finite sum, we use an integral that considers every possible $t \in [0, T]$. Based on what has been derived in Question 1, we know that if death occurs at time t , the resulting discounted expected cash flow has value $C_0 e^{(g-r)t} + \frac{C_0}{S_0} \mathbb{E}^Q[e^{-rt} (S_t - S_0 e^{gt})_+] | \mathcal{F}_0$. The probability density function of dying at time $\tau = t$ is given by $f(t) = {}_t p_x \mu_{x+t}$, where the force of mortality μ_{x+t} is such that ${}_t p_x = \exp(-(\int_0^t \mu_{x+s} ds))$.

Therefore, the value of the GMDB part of the contract is given by

$$\begin{aligned} V_0^{GMDB} &= \int_0^T f(t) \mathbb{E}^Q[e^{-rt} C_0 \max(\frac{S_t}{S_0}, e^{gt}) | \mathcal{F}_0] dt \\ &= \int_0^T {}_t p_x \mu_{x+t} (C_0 e^{(g-r)t} + \frac{C_0}{S_0} \mathbb{E}^Q[e^{-rt} (S_t - S_0 e^{gt})_+ | \mathcal{F}_0]) dt \end{aligned}$$

The embedded call options with strike $K(t) := S_0 e^{gt}$ are priced with Black-Scholes:

$$\mathbb{E}^Q[e^{-rt} (S_t - K(t))_+ | \mathcal{F}_0] = S_0 \Phi(d_1^{BS}(t)) - K(t) e^{-rt} \Phi(d_2^{BS}(t)) .$$

Finally, the total contract value is the sum of both components:

$$V_0 = V_0^{GMAB} + V_0^{GMDB} .$$

Question 2

Binomial tree implementation

The contract is valued using a discrete-time binomial tree that approximates the risk-neutral dynamics of the underlying asset $(S_t)_{t \geq 0}$. Time to maturity T is divided into N equal time steps $\Delta t = T/N$. At each step, the price can move up or down:

$$S_{t+\Delta t} = \begin{cases} S_t \cdot u & \text{with probability } p \\ S_t \cdot d & \text{with probability } 1 - p \end{cases}$$

where the up and down factors are defined as in the lecture:

$$u = e^{(r - \frac{\sigma^2}{2})\Delta t + \sigma\sqrt{\Delta t}} \quad \text{and} \quad d = e^{(r - \frac{\sigma^2}{2})\Delta t - \sigma\sqrt{\Delta t}}.$$

The associated risk-neutral probability is $p = \frac{e^{r\Delta t} - d}{u - d}$.

The underlying asset value S_t is calculated at each node so as to account for all price paths at every time step. The asset value at step $i \in \{0, \dots, N\}$ after $j \in \{0, \dots, i\}$ upward moves, denoted S_i^j , is given by

$$S_i^j = S_0 u^j d^{i-j}.$$

To initiate the tree, the contract payoffs are first calculated at terminal nodes, corresponding to the payoffs at maturity T . For these nodes, the contract value is

$$V_N^j = C_0 \max\left(\frac{S_N^j}{S_0}, e^{gN}\right).$$

Then, the contract value is computed backward in time. In the case of survival during the considered interval, the expected "continuation" payoff is calculated with the following formula:

$$\text{continuation_payoff} = e^{-r\Delta t}(pV_{i+1}^{j+1} + (1-p)V_{i+1}^j).$$

The mortality risk is introduced via the force of mortality parameter μ_x . For simplicity, we can assume a constant rate $\mu_x = \mu$ such that the probability of survival during a chosen interval $[t, t + \Delta t]$ is $\Delta t p_{x+t} = Q(\tau > \Delta t) = \exp(-\int_0^{\Delta t} \mu_{x+s} ds) = e^{-\mu\Delta t}$ and the probability of death over $[t, t + \Delta t]$ is $1 - e^{-\mu\Delta t}$.

In case of premature death during the interval $[t, t + \Delta t]$, the insurer pays immediately $C_0 \max(\frac{S_t}{S_0}, e^{gt})$ (approximation that becomes negligible as $\Delta t \rightarrow 0$), so

$$\text{death_payoff} = C_0 \max\left(\frac{S_i^j}{S_0}, e^{g(i\cdot\Delta t)}\right).$$

We finally merge these payoffs thanks to a probability-weighted average:

$$V_i^j = e^{-\mu\Delta t} \text{continuation_payoff} + (1 - e^{-\mu\Delta t}) \text{death_payoff}.$$

The backward induction is carried out until the contract value V_0^0 is obtained.

Numerical valuation with analytical formula

In order to assess the precision of the binomial tree values, we want to compute the real contract numerical value. The GMAB component can be found without much difficulty by applying the formulas derived in Question 1. We find

$$K = 100e^{0.025 \cdot 5} = 113.3149,$$

$$d_1^{BS} = \frac{\ln \frac{100}{113.3149} + (0.032 + \frac{0.12^2}{2}) \cdot 5}{0.12\sqrt{5}} = 0.2646 \quad , \quad d_2^{BS} = 0.2646 - 0.12\sqrt{5} = -0.0037 \quad ,$$

$$\Phi(d_1^{BS}) = 0.6043 \quad , \quad \Phi(d_2^{BS}) = 0.4985 \quad ,$$

$$\mathbb{E}^Q[e^{-rT}(S_T - K)_+ | \mathcal{F}_0] = 100 \cdot 0.6043 - 113.3149e^{-0.032 \cdot 5} \cdot 0.4985 = 12.2945 \quad ,$$

$$V_0^{GMAB} = {}_T p_x (e^{(0.025 - 0.032) \cdot 5} \cdot 100000 + \frac{100000}{100} \cdot 12.2945) = {}_T p_x \cdot 108855.0416 \quad .$$

Assuming a force of mortality $\mu_x = \mu = 2\%$, the probability of survival is ${}_T p_x = e^{-0.02 \cdot 5} = 0.9048$. Therefore, the GMAB component has a value

$$V_0^{GMAB} = 0.9048 \cdot 108855.0416 = 98498.7970 \quad .$$

The evaluation of the GMDB part involves the computation of an integral, which can be approached with a numerical approximation. Indeed, from Question 1, the value of the death benefit is given by

$$V_0^{GMDB} = \int_0^T {}_t p_x \mu_{x+t} (C_0 e^{(g-r)t} + \frac{C_0}{S_0} \mathbb{E}^Q[e^{-rt}(S_t - S_0 e^{gt})_+ | \mathcal{F}_0]) dt \quad .$$

In order to compute this integral numerically, we discretize the interval $[0, T]$ into N subintervals of size $\Delta t = T/N$. The integral is then approximated by a Riemann sum:

$$V_0^{GMDB} \approx \sum_{i=1}^N \left(C_0 e^{(g-r)t_i} + \frac{C_0}{S_0} \mathbb{E}^Q[e^{-rt_i}(S_{t_i} - S_0 e^{gt_i})_+ | \mathcal{F}_0] \right) Q(t_{i-1} \leq \tau < t_i),$$

where $t_i = i \cdot \Delta t$, and the probability of death over the interval $[t_{i-1}, t_i]$ is $Q(t_{i-1} < \tau \leq t_i) = Q(\tau > t_{i-1}) - Q(\tau > t_i) = {}_{t_{i-1}} p_x - {}_{t_i} p_x = e^{-\mu t_{i-1}} - e^{-\mu t_i}$. With $N = 10^6$, this gives us

$$V_0^{GMDB} \approx 10102.8930 \quad .$$

The numerical value of the contract is thus

$$V_0 = 98498.7970 + 10102.8930 = 108601.6900 \quad .$$

Convergence of binomial tree prices

To validate the numerical accuracy of the binomial tree approach, we study the convergence of the contract value as the number of time steps N increases.

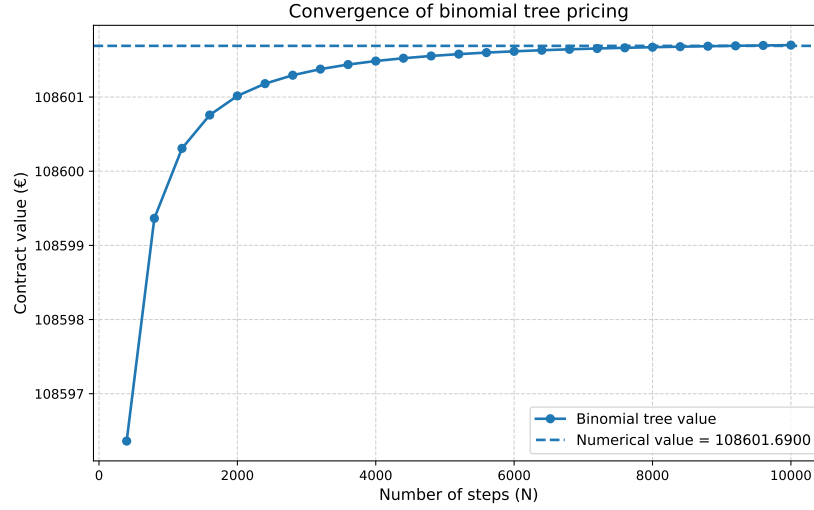


Figure 1: Convergence of the binomial tree prices.

As Figure 1 shows, the approximation error is significant for small values of N due to the coarse discretization of time, and as N increases, the prices of the binomial tree converge towards the reference numerical value (108601.6900€). This shows that the binomial model converges towards Black-Scholes continuous-time valuation model.

Question 3

Monte-Carlo implementation

We now value the contract using Monte-Carlo simulations in order to validate the prices obtained with the binomial tree. As outlined earlier, the dynamics of $(S_t)_{t \geq 0}$ is a geometric diffusion under the risk-neutral measure Q :

$$\frac{dS_t}{S_t} = rdt + \sigma dW_t .$$

The process is discretized over N time steps of size $\Delta t = T/N$, leading to the Euler discretization scheme:

$$\Delta S_{j\Delta t} = S_{(j-1)\Delta t}(r\Delta t + \sigma\sqrt{\Delta t}\epsilon) \quad , \quad S_{j\Delta t} = S_{(j-1)\Delta t} + \Delta S_{j\Delta t} \quad \text{for } j \in \{1, \dots, N\} .$$

where $\epsilon \sim \mathcal{N}(0, 1)$.

As required, mortality is not explicitly simulated. Instead, expected discounted cash-flows are weighted by survival and death probabilities. Assuming again a constant force of mortality μ , the survival probability is ${}_t p_x = e^{-\mu t}$ and the probability of death during the interval $[t_{i-1}, t_i]$ is $Q(t_{i-1} < \tau \leq t_i) = {}_{t_{i-1}} p_x - {}_{t_i} p_x$, where $t_i = i \cdot \Delta t$.

For k th sample path $(S_{t_j}^{(k)})_{j=0, \dots, N}$, the discounted payoff at time t_i is $e^{-rt_i} C_0 \max(\frac{S_{t_i}^{(k)}}{S_0}, e^{gt_i})$, so that the value of the contract for the k th simulation is given by

$$V_0^{(k)} = \sum_{i=1}^N e^{-rt_i} C_0 \max(\frac{S_{t_i}^{(k)}}{S_0}, e^{gt_i}) Q(t_{i-1} < \tau \leq t_i) + e^{-rT} C_0 \max(\frac{S_T^{(k)}}{S_0}, e^{gT}) {}_T p_x .$$

The final value of the contract is obtained by averaging over M simulated paths:

$$V_0 = \frac{1}{M} \sum_{k=1}^M V_0^{(k)} .$$

Convergence of the Monte-Carlo prices

To assess the convergence of the Monte-Carlo estimator, the contract value is computed for increasing numbers of simulations M . For each value of M , the estimation is repeated several times in order to compute both the mean and the standard deviation.

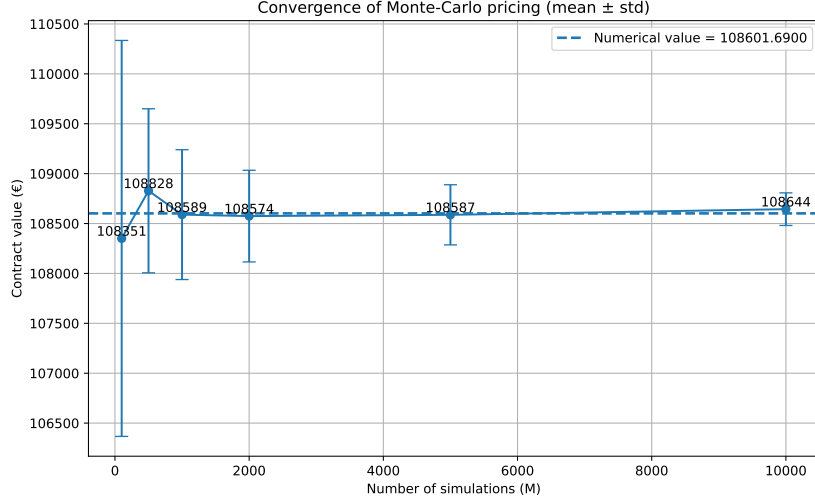


Figure 2: Convergence of the Monte-Carlo prices. The number of discretization steps was set to $N = 1000$.

Figure 2 presents those means and standard deviations, along with the benchmark numerical value. As expected, the Monte-Carlo estimator converges towards the analytical value as the number of simulations increases, and the variance of the estimator decreases with M , illustrated by the shrinking error bars. We also observe that the estimation error decreases in $\mathcal{O}(\frac{1}{\sqrt{M}})$, in line with the theoretical convergence of Monte-Carlo methods.

The Monte-Carlo results are hence consistent with both the binomial tree valuation and the actual price obtained numerically with the analytical formula. This confirms the validity of all three approaches